Gauresh Rangaiyan

College Station, TX | (409)-888-2711 | gaureshrangoo@gmail.com | LinkedIn | Portfolio | Personal Website

EDUCATION

Master of Science, Mechanical Engineering **May 2028**
Texas A&M University *College Station, TX*
• **Honors/Awards:** Brenda & Jerry Gray '62 Fellowship (2026–2027)

Bachelor of Science, Aerospace Engineering **May 2026**
Texas A&M University | Minor: Mathematics | GPA: 3.55 *College Station, TX*
• **Honors/Awards:** Distinguished Student (3.75+ GPA) (Spring 2026)

EXPERIENCE

Mechanical Engineer Thesis Researcher **Sept 2026 - Present**
Texas A&M University *College Station, TX*
• **Designing 2-DOF** vibration platform for testing scaled planetary landing footpads under simulated landing conditions
• **Developing SolidWorks** models and **3D-printed** vibration-platform components with integrated linear guides for vibration platform
• **Integrating 12–24V DC vibration motor** to generate mechanical excitation for preliminary vibration testing and characterization
• **Supporting** experimental test setup, simulant preparation, transducer instrumentation, and mechanical data collection to evaluate landing-footpad response under simulated planetary surface conditions

Mechanical Engineering Intern **June 2026 - Aug 2026**
Pioneer Design & Engineering Pvt Ltd *India*
• **Evaluated** satellite wall panels using **SolidWorks**, optimizing design to increase **strength-to-weight** ratio by **22%** and reduce mass **28%**
• **Cut** panel mass by **35%** and simplified production by re-engineering **CNC-machinable** integral-skin structure, eliminating bonded layers
• **Validated ANSYS simulations** against physical testing data within **5.4–13.9% deviation** under **11G triaxial launch loads**
• **Designed** compliant aviation brackets and structural levers by applying **DFM**, **prototyping**, and cross-functional **collaboration**

Aerospace Engineering Researcher **Aug 2024 - Dec 2024**
Texas A&M University *College Station, TX*
• **Achieved** up to **10x** increase in energy absorption in a foam composite material by engineering a novel reinforcement approach that embeds precision-printed structural elements inside foam using a modified **3D printer** platform
• **Performed structured DOE** varying geometry, increasing energy absorption performance by **91% (nearly 3x)** and identifying most efficient design configuration by fabricating **PMC-780 elastomer specimens** with multiple **3D-printed** molds and fixtures
• **Evaluated** composite performance through compression testing and **Python-based data analysis,** demonstrating **1.7–2.9×** greater force resistance than independent components and only **4%** additional degradation over **9 stress cycles**

Mechanical Engineering Researcher **May 2024 - Aug 2024**
Indian Institute of Technology *India*
• **Modeled SMA-actuated soft robotic finger** in **SolidWorks**, integrating resistance wiring, internal channels, joints, and load-transfer mechanisms, producing parts within **5%** of **CAD dimensions** for reliable assembly and repeatable motion
• **Iterated mechanical architecture** by refining **SMA** placement, routing, pretension, geometry, and structural interfaces, enabling **flexion of 100 degrees** in approximately **4 seconds** and passive return to resting position in **13 seconds**
• **Performed root-cause analysis** on asymmetric bending through repeated actuation testing, diagnosing channel misalignment and adjusting internal geometry, decreasing offset from **0.3–0.4 mm** to **0.1 mm**, improving surface locomotion and durability
• **Built** and **validated** functional prototypes end-to-end through **3D-printed fabrication, silicone casting, SMA/resistance-wire installation** and **mechanical assembly,** achieving a peak fingertip force of **6.5 N** under an **8 V supply**

PROJECTS

Lab-Assisting Robotic Manipulator (LARM) | SolidWorks, FEA, Rapid Prototyping, 3D Printing
• **Implemented end effector** of **6-DOF robotic arm**, discussing multiple concepts and comparing performance **under 50N** representative gripping loads through **FEA** to balance **structural integrity with manufacturability**
• **Collaborated** with a multidisciplinary engineering team to develop concepts for wet-lab tasks, refining end-effector geometry and mechanical interfaces to support precise, reliable gripping and manipulation, ensuring positioning accuracy within **±0.5 mm**

Planetary Gearbox |SolidWorks, Rapid Prototyping, 3D Printing
• **Prototyped** in **SolidWorks** a **5:1 planetary gearbox** using **module 1.5 gears** to maximize **torque/motion** and minimize **backlash**
• **Devised** and **assembled** a compact **3D-printed gearbox** with integrated bearings, achieving smooth transmission and high torque output

SKILLS

Technical: SOLIDWORKS, Fusion 360, Onshape, GD&T, Python, FEA (ANSYS), DFM, 3D Printing, CNC Mill/Turn, G-code, Mastercam